CityUHK-Style Beamer Template

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Innovating into the Future

Outline

1. Core Layout
2. Figure and Explanation
3. Pure Math
4. Derivation Logic
5. Summary

Core Layout

Design Principles

What the template is optimized for

Built-in page types

- Title page with strong visual hierarchy
- Automatic section-divider slides
- Regular content slides with a clean header
- Dedicated ending slide for Q&A or wrap-up

Navigation

The left sidebar keeps the section map visible throughout the talk and highlights the current section.

Use cases

Thesis defenses, conference talks, lecture notes, progress reports.

How to Reuse the Template

Minimal author workflow

1. Update the metadata in the preamble: title, author, institute, and date.
2. Organize the talk using `\section{...}` so the sidebar stays informative.
3. Replace the demo frames with your own figures, equations, and tables.
4. End with `\CityUHKClosingFrame{...}{...}`.

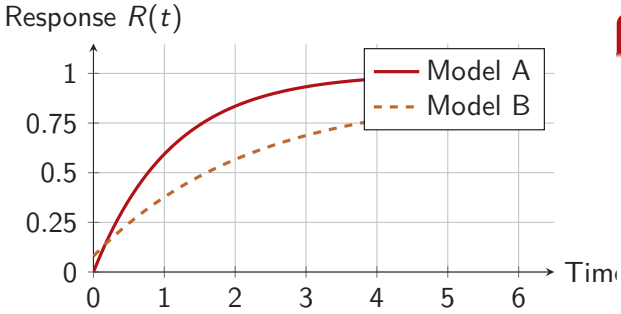
File structure

`beamerthemeCityUHK.sty` contains the visual theme and `cityu_beamer_example.tex` is the editable example deck.

Figure and Explanation

Figure with Explanation

A common research-slide pattern



Reading the figure

- Model A converges faster to the steady state.
- Model B has a slower transient but a slightly elevated baseline.
- The visual contrast makes the main comparison immediately visible.

Presentation tip

Pure Math

Pure Math Example

Compact but readable notation

Variational problem

For $u \in H_0^1(\Omega)$, consider the energy functional

$$\mathcal{E}(u) = \int_\Omega \left(\frac{1}{2} |\nabla u|^2 - fu \right) \, \mathrm{d}x.$$

Euler–Lagrange equation

If $u^\star$ minimizes $\mathcal{E}$, then for every test function $\varphi \in H_0^1(\Omega)$,

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} \mathcal{E}(u^\star + \varepsilon\varphi) \Big|_{\varepsilon=0} = \int_\Omega \nabla u^\star \cdot \nabla \varphi - f\varphi \, \mathrm{d}x = 0.$$

Interpretation

Derivation Logic

Derivation Logic Example

Show the chain of reasoning, not only the result

Goal

Derive the closed-form least-squares estimator for

$$y = X\beta + \varepsilon, \qquad X \in \mathbb{R}^{n\times p}.$$

1. Define the objective
$$J(\beta) = \|y - X\beta\|_2^2 = (y - X\beta)^\top (y - X\beta).$$
2. Expand and differentiate with respect to β
$$\nabla_\beta J(\beta) = -2X^\top y + 2X^\top X\beta.$$
3. Set the gradient to zero to obtain the normal equation
$$X^\top X\beta = X^\top y.$$

Derivation Logic Example

Useful framing for spoken explanation

What the audience needs

- The objective function
- The key algebraic manipulation
- The condition that justifies the final step

What can stay implicit

- Routine matrix-calculus details
- Intermediate terms that cancel
- Side cases deferred to backup slides

Speaking note

A derivation slide works best when each displayed equation advances exactly one logical step.

Summary

Summary

What this template already gives you

- A CityUHK-like visual direction with dark red, gold, and neutral accents
- A persistent sidebar navigation panel with current-section highlighting
- Automatic section pages and a dedicated closing slide
- Example layouts for technical talks with figures and mathematics

Next step

Replace the sample content with your own title, sections, data visualizations, and equations.

Thank You

Questions and discussion